

Module 08: Planning — The Robot Mission

Learning Objectives

1. Understand that **Planning** is thinking ahead.
2. Learn about **Path Planning** (Finding the best way).
3. Know that robots must plan to avoid obstacles.

Introduction: The Robot Map

Imagine you are a robot in a maze. You want to get to the Charger (Goal). There is a wall in the way! What do you do? You make a **Plan**.

- "If I go straight -> Crash."
- "If I turn Left -> Clear path." The robot imagines the path *before* it moves.

Key Concepts

1. The Map

To plan, a robot needs a map in its head. It marks:

- **Open Space** (Safe).
- **Obstacles** (Walls, Tables, Cats).
- **The Goal** (Where to go).

2. Shortest Path

Robots are smart. They don't want to walk in circles. They calculate the **Shortest Path**. Why walk 100 steps if you can walk 10? This saves battery!

3. Re-Planning

Uh oh! A chair moved! The map is wrong! The robot must **Re-Plan**. "New Obstacle detected. Calculating new route..."

Activities

Activity 1: Robot Maze

Draw a maze on the floor with tape. One person is the **Controller**. One person is the **Robot**. The Controller must give a full list of commands *before* the Robot moves. "Forward 3. Left Turn. Forward 2." If the robot crashes, the Plan failed!

Activity 2: Obstacle Course

Build a course with pillows. Program a toy robot (or a friend) to get through it. Can you find the shortest path?

Summary

Planning is how robots survive in a changing world. They use maps and math to find the best way to their goal without crashing!

References

- *Maps and Globes* by Jack Knowlton
- *Coding with Anna and Elsa* (Frozen coding book about paths)